
Hidden States Turn Chain-of-Thought Tokens into Working Memory

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Abstract

1 Where do language models store the intermediate values used in multi-step reason-
2 ing? One hypothesis is that models reason in an internal workspace and write
3 chain of thought only when that workspace becomes insufficient. We find instead
4 that the written positions themselves hold the state. Holding a model’s reasoning
5 text fixed to the character, we overwrite the hidden state at one written number
6 with the state from a run that wrote a different number: the final answer follows
7 the never-written number in 70-100% of runs on six off-the-shelf models from
8 four families (4B to 14B), size-matched random controls reach at most 1% on the
9 four models without reasoning distillation, and on Qwen3-4B the planted state
10 stays causally effective across 40 subsequent reasoning steps. Later computation
11 retrieves the stored value by attending to its position (blocking that attention re-
12 moves the effect), and two positions hold independently settable values that com-
13 pose into a sum appearing nowhere in the text (88-90% of runs). The capacity-
14 triggered account finds no support: forbidden to write, models from 1.5B to 671B
15 parameters collapse beyond one or two dependent steps, yet writing shows no on-
16 set threshold (93-100% of intermediate values written from depth one), and no
17 intervention, including ablating the selected top-16 concept directions of J-space,
18 a recently proposed residual-stream workspace, moves state between the trace and
19 internal stores. For oversight, an identical visible reasoning prefix can carry dif-
20 ferent hidden states that lead to different answers; whether such divergences arise
21 without intervention is open.

22 1 Introduction

23 A language model doing multi-step reasoning computes a value, writes it into its chain of thought,
24 and uses it several steps later. The model could maintain the value independently of its output, or
25 bind it to the emitted token position and retrieve it through attention. For serial numeric reasoning
26 we give a causal answer: the state is bound to the written position. We call this trace-indexed
27 memory, task state attached to an emitted token position and read back by attending to that position,
28 as opposed to trace-independent memory. It qualifies as working memory operationally: the state
29 persists at earlier positions, is selectively retrieved during later computation, and can be manipulated
30 independently across positions.

31 Among items where the model demonstrably relies on the written intermediate (Figure 1A), the
32 planted value redirects the final answer in 0.70 to 1.00 of runs across six off-the-shelf models from
33 four families; size-matched random noise redirects at most 0.01 (Section 4). On Qwen3-4B the
34 planted state remains causally effective across 40 subsequent reasoning steps.

35 Four experiments establish the mechanism:

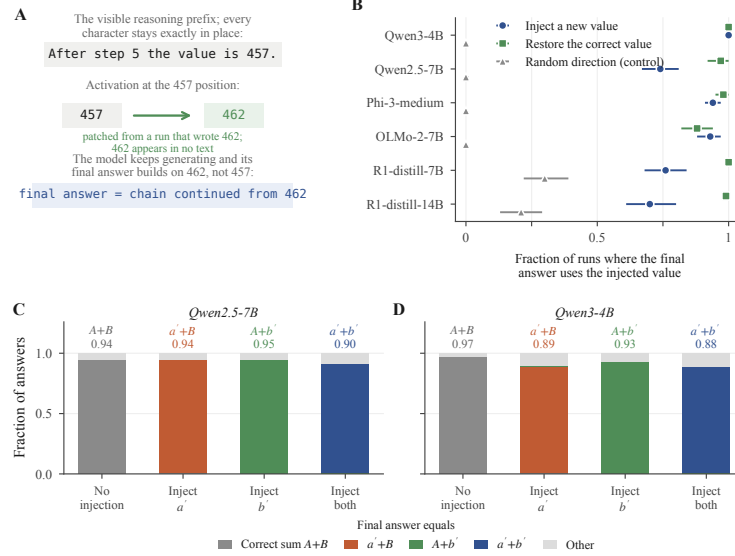


Figure 1: Same visible prefix, different stored value, different answer. (A) The reasoning prefix ends “After step 5 the value is 457.” The text never changes; the activation at the 457 position is replaced with one captured from a run that wrote 462, and the final answer continues from 462, a value appearing nowhere in any text. (B) Fraction of runs whose final answer uses the injected value, per model: planting a counterfactual value p (circles), restoring the correct value c under edited text (squares), and a size-matched random control (triangles); bars are 95% bootstrap intervals (the distilled models’ elevated random control is explained in Appendix A). (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

1. **Write** (behavioral, 1.5B to 671B). Models externalize intermediate values from the first step at every scale, with no onset threshold; denied writing they fail beyond one to two dependent steps, and no intervention shifts state across the boundary (Section 3).
 2. **Store** (causal, 4B to 14B). The hidden state at a written value contains causally active task state: restoring it corrects a corrupted trace, and planting a counterfactual value redirects the answer (Section 4).
 3. **Retrieve** (causal, 4B to 14B). Later computation accesses the stored value by attending to its position; blocking that attention removes the dependence, and layer sweeps place the read in the middle third of the network (Section 5).
 4. **Compose** (causal, 4B and 7B). Two positions hold independently settable values that combine into a sum absent from all preceding text (Section 6).
- The contrasting account centers on J-space, a proposed global workspace identified by a Jacobian-based lens over residual activations [Gurnee et al., 2026], and it makes two separable predictions. A capacity-triggered workspace predicts writing begins at the depth where silent performance fails; writing instead begins at step one at every scale, and no intervention induces the predicted shift between internal and written state (Section 3). A causally load-bearing workspace predicts that removing its directions impairs reasoning; ablating the selected top-16 J-space concept directions instead has no greater effect than ablating size-matched random directions at any tested strength, while the planted value sits in the same residual states the lens reads (Section 3). Causally important task state need not be localized to the concept directions a fitted lens selects.
- Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on program-variable tasks [Zhu et al., 2025]; the contribution here is a causal account of storage, retrieval, and composition on off-the-shelf models.

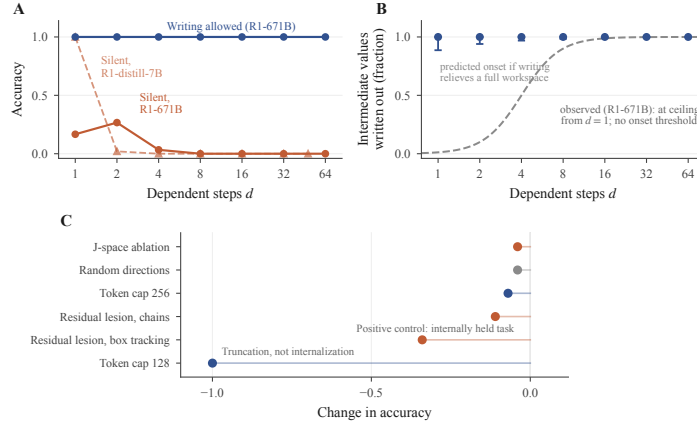


Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 0.93 to 1.00 at every depth (Wilson intervals); the dashed curve is the onset a capacity-relief account predicts, and it does not appear. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task far more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

2 Experimental setup

2.1 Tasks and models

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count of an item changes through varied events and surface forms. In both, the model writes one line per step giving the running value. A third task, box tracking, updates several containers in parallel; it appears only as the internally held comparison for the lesion test. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; the 1.5B distill appears in the behavioral scale range only. Frontier models (DeepSeek V3.2, DeepSeek-R1-671B, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

2.2 Edit-and-continue protocol

The protocol takes a correct worked solution, edits the value at one step, truncates immediately after the edited line, and lets the model continue. Three values recur: the correct c from the original solution, the edited e shown in the text, and a counterfactual p whose activation is planted. The conditions: a *text edit* shows e with no state intervention; *restore* shows e with the position’s hidden state restored to c ’s; *plant* shows e with the state replaced by p ’s, captured from a run that wrote p ; the *random control* shows e with a size-matched random perturbation. The patched state is the residual-stream activation at the value’s position across a middle band of layers; interventions cover every token of multi-token values (Appendix C). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95% bootstrap intervals over items.

83 3 Write: no inference-time capacity threshold

84 3.1 Performance without written intermediates

85 Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else
86 was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at
87 $d = 4$; DeepSeek-R1-671B: at most 0.27 at any depth). Inert filler tokens in place of a trace [Pfau
88 et al., 2024] do not rescue the collapse (Appendix C). The values are random, so nothing can be
89 recalled; the failures bound performance under this instruction, with its prompting and training-
90 distribution effects, rather than latent internal capacity.

91 3.2 When models write intermediate values

92 Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through
93 $d = 64$, at every scale tested: no difficulty threshold at which writing begins is observed. This
94 rules out an inference-time switch from internal holding to writing; an always-write policy learned
95 in training could itself descend from capacity limits.

96 3.3 Tests for reallocation between internal and written state

97 A residual-stream lesion at fixed dose lowers accuracy on box tracking, whose state is held inter-
98 nally, by 0.34, and on the written chain task by 0.11; a lesion strong enough to cross the accuracy
99 cliff makes traces disintegrate rather than expand. Hard token budgets compress wording up to 2.5
100 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step.
101 Ablating the selected top-16 J-space concept directions at every position has no greater effect than
102 ablating size-matched random directions: at a calibrated generation-preserving dose it leaves chain-
103 of-thought accuracy, bare-answer accuracy, and the amount written unchanged on synthetic chains
104 and on GSM8K, and a dose sweep from a quarter of that strength to complete removal of the selected
105 directions keeps accuracy at 0.99 to 1.00 on one- and two-step chains (Appendix C). The chain-of-
106 thought-versus-direct ablation asymmetry reported for J-space [Gurnee et al., 2026] therefore does
107 not replicate on our tasks at any tested strength, including the depths where models do perform the
108 arithmetic internally. Section 4 shows a causally decisive planted value sitting in the same residual
109 states the lens reads.

110 4 Store: written positions hold causally active values

111 Figure 1 decomposes the effect of editing a written value, conditioned on the cases where the mem-
112 ory is demonstrably in use: items on which the model follows the visible text edit to e (0.44 to 0.98
113 by model; Appendix A). Among these, restoring c ’s state while the text still shows e returns the
114 answer to correct in 0.88 to 1.00 of cases, and planting p ’s state redirects the answer to p , a value
115 appearing nowhere in any text, in 0.70 to 1.00 of cases. The size-matched random control sits at
116 or below 0.01 for the four models without reasoning distillation. The patch replaces the position’s
117 full contextual activation; it shows that state is causally sufficient to make later computation be-
118 have as if p had been written, and it does not isolate a numeric feature within that state. Because
119 p is a mid-chain intermediate the model never produced, the patch does not inject the answer. The
120 effect persists to 40-step chains (0.95 to 0.98 on Qwen3-4B) and replicates on the narrative task
121 (Appendix A).

122 5 Retrieve: attention to the written position

123 5.1 Attention-blocking experiments

124 Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s
125 dependence on that value on every model tested (Figure 3A); matched blocks of the adjacent words,
126 a random prompt position, or a random trace position leave it at the unblocked level. Blocking
127 the operand the next step needs breaks the arithmetic itself, a positive control for the mask. The
128 collapse holds across three families and both task formats (natural-language: 0.730 to 0.007, 0.720

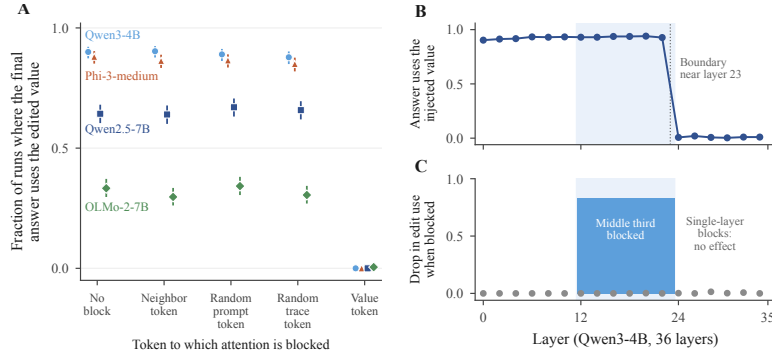


Figure 3: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to near zero on four models; matched control blocks change nothing. Points are means with 95% Wilson intervals ($n=600$ runs per condition). (B) Overwriting the hidden state at any single layer before the read redirects the answer; after it, nothing (Qwen3-4B). (C) Blocking attention across the shaded middle band removes most edit use; no single layer’s block matters (gray dots).

129 to 0.003). Under the block the model continues fluently, with answers matching neither the edited
 130 nor the correct value (unparseable rate 0.000; OLMo-2-7B 0.040, its unblocked level).

131 5.2 Where retrieval occurs across layers

132 Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers;
 133 Figure 3B, C). Overwriting the value’s hidden state at any single layer up to 22 redirects the
 134 answer at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten
 135 at any layer before the read and at none after. Blocking the value-position attention only at the
 136 middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the
 137 early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is
 138 distributed redundantly across the middle third and confined to it; the two sweeps agree on the same
 139 boundary, near layer 23. A coarse sweep on Phi-3-medium shows the same structure; Qwen2.5-7B
 140 is unresolved at this granularity (Appendix B).

141 6 Compose: independent storage across positions

142 In a task with two counters summed at the end (Figure 1C, D), planting a never-written value at one
 143 final position yields the corresponding mixed sum (0.94 and 0.95 on Qwen2.5-7B, 0.89 and 0.93 on
 144 Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88 (products of the single rates: 0.89
 145 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95). The reported number
 146 must be assembled at answer time from two values set independently and never written anywhere.

147 7 When models retrieve versus recompute

148 By default, models reuse the value they wrote rather than recompute it, and what triggers recom-
 149 putation is model-specific: DeepSeek V3.2 reverts an edit when one sentence in the prompt states
 150 the value’s derivation (0.43 following, against 0.93 when the prompt only implies the value), Claude
 151 Sonnet 4.5 when the edited step is the value’s only source (0.22 against 0.81), while Llama 3.3 70B
 152 follows edits at 0.97 despite solving the same chains from scratch, and the Qwen models’ following
 153 rates in Table 1 leave little room for reversion (conditions, controls, and cell sizes in Appendix C).
 154 On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10
 155 by V3.2, and within one model the contrast holds, V3.2’s 0.93 on chains against 0.10 on GSM8K:
 156 a benchmark intermediate is often one operation from numbers still in the prompt. Lanham et al.
 157 [2023] report reliance on written reasoning declining with scale on standard benchmarks; our con-
 158 trasts offer a candidate factor, the regenerability of the state, which influences reliance without de-
 159 termining it, and neither scale nor recomputation ability alone predicts it. Trace reliance is separate

160 from verbalization faithfulness, whether the trace reports the computation behind the answer; our
161 claims concern reliance on already-written state.

162 8 Related work

163 Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal
164 abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches
165 conflate a component mattering with deleting information at a position, which the single-layer
166 sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and
167 filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate
168 internally. Wang [2026] corrupt surface traces only in aggregate and conclude reasoning is latent;
169 our token-level edits show written state is causally decisive when the model cannot regenerate it.
170 Korbak et al. [2025] argue chain-of-thought monitoring is a useful and fragile oversight channel;
171 our results bound what such a monitor observes.

172 9 Limitations

173 Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans, and
174 richer multi-entity state are untested. The J-space null bounds the fitted readout and the selected
175 active directions, on our tasks and doses: the full-strength sweep covers six middle layers of one
176 model, and the lens’s domain excludes attention and MLP interiors, the KV cache, and residual
177 directions outside the fitted dictionary. The fine layer sweeps cover one model. White-box evidence
178 is at 4B to 14B scale, frontier evidence behavioral. Conditioning on edit-following makes cross-
179 model levels not directly comparable (Appendix A). A structured patch control planting a valid
180 activation for another quantity would sharpen the value-content claim.

181 10 Conclusion

182 Written positions function as persistent working memory in the serial reasoning tasks we study:
183 their hidden state can be changed independently of the visible prefix, is retrieved through middle-
184 layer attention, and composes across positions to determine later answers. A monitor that reads
185 the completed trace sees the text; under intervention, identical visible prefixes lead to different
186 subsequent reasoning and answers [Korbak et al., 2025]. Whether comparable divergences arise
187 naturally, and in richer reasoning than numbers, remains open.

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213 A Per-model edit and patch rates

214 Table 1 gives the per-model rates behind Figure 1B, with the text-edit-following rate that conditions
 215 the patch columns. Ten-step chains; brackets are 95% bootstrap intervals; n counts items where the
 216 text edit was followed. Unconditioned per-run outcomes are in the released data. The distill models
 217 re-solve the problem after their reasoning phase, which elevates their random-control reversion: in
 218 a text-logged sample, 24 of 24 reverting random-control continuations re-derive from the prompt’s
 219 starting value after the reasoning phase, and none correct the value in place. Re-solving can only
 220 produce the correct answer, so the planted-value rates are unaffected. R1-distill-1.5B is excluded
 221 (12 usable items; random control exceeds the planted-value rate). On the narrative task the pattern
 222 replicates: a planted never-written count is followed at 0.83 and 0.86, restoring the correct state
 223 returns the answer at 0.89 and 0.94, and the random control is 0.00, on Qwen3-4B and Phi-3-medium
 224 ($n = 86, 89$).

Table 1: Text-edit following and state-patch outcomes by model.

Model	Text edit followed	Restore correct state	Plant never-written value	Random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

225 B Coarse cross-model layer sweeps

226 The state patch applied to one third of layers at a time ($n = 60$ items per model; fraction of runs
 227 following the planted value). Qwen3-4B (36 layers): early 0.91, middle 0.93, late 0.00, all layers
 228 0.92. Phi-3-medium (40 layers): early 0.89, middle 0.87, late 0.01, all layers 0.89. Both match the
 229 fine sweep’s structure: the planted value takes effect at any band before the read and at none after.
 230 Qwen2.5-7B (28 layers) follows the planted value at 0.70 to 0.73 in every band, so its read boundary
 231 is not resolved at this granularity.

232 Reading the written value is the preferred path, with reconstruction from earlier lines a model-
 233 specific fallback: with the value’s characters replaced by unreadable dots and attention untouched,
 234 the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60
 235 to 0.73 of cases; Phi-3-medium never does (0.00).

236 C Generation and intervention configuration

237 Generation uses temperature 0.6, top- p 0.95, five samples per item. The text edit changes only
 238 the visible text of the value. A written value may span several tokens; the state patch and the
 239 attention block apply to every token of the value’s span. The state patch overwrites the residual
 240 stream at those positions across a middle band of layers with activations captured from a run in
 241 which the counterfactual value was written there; the attention block masks all attention from post-
 242 edit positions to those positions at all layers. White-box sample sizes are 100 to 600 per condition
 243 with two to three seeds; the behavioral read-versus-recompute experiments of Section 7 contain
 244 1,300 to 2,400 scored runs per cell.

245 The filler condition prefills the assistant turn with twelve inert dot tokens per dependent step, then
 246 decodes the answer at temperature 0. Full cells ($n = 30$ each): at $d = 2$, 0.97 with filler against
 247 0.40 without on Qwen3-4B and 0.53 against 0.47 on Qwen2.5-7B; at $d = 4$, 0.07 against 0.00 and
 248 0.13 against 0.07; at $d = 8$, 0.00 for both models in both conditions. Qwen3-4B’s recovery at $d = 2$

249 is the parallel-width benefit filler tokens are known to provide, and it does not extend to the depths
250 where writing is needed.

251 In the DeepSeek V3.2 evidence conditions of Section 7, with the derivation-stating sentence present
252 but the edit two steps downstream, following returns to 0.96, so the reversion targets the stated value.
253 The evidence conditions are separate experiments; we claim within-experiment contrasts only.

254 The J-space ablation dose is calibrated on neutral text to the strongest strength preserving ordinary
255 generation (next-token agreement 0.87, perplexity ratio 1.12). The dose sweep runs the ablation of
256 Section 3 at $k = 16$ and $\alpha \in \{0.125, 0.25, 0.5, 0.75, 1.0\}$ on layers 9 to 14 of Qwen2.5-7B with
257 identical items across conditions; accuracy stays 0.99 to 1.00 in every condition on one- and two-
258 step chains ($n = 80$ per cell). Prompts, model revisions, and per-run configuration files are released
259 with the code and data.